

Statistical Calculations

- Statistics: **Variance**
- Estimated **variance** of attribute j : $\hat{\sigma}_j^2 = \frac{1}{n-1} \cdot \sum_{i=1}^n (x_{ij} - \hat{\mu}_j)^2$
- Statistics: **Total Variance**
- What is the total **variance** in a numerical data set?
- $Var(D) = \hat{\sigma}_1^2 + \hat{\sigma}_2^2 + \hat{\sigma}_3^2 + \dots + \hat{\sigma}_n^2$
- Statistics: Estimated **Covariance**
- What is the estimated **covariance** between two attributes in a numerical data set?
- Ex: Estimated **Covariance** between attribute 1 and 2?

$$\hat{\sigma}_{12} = \frac{1}{n-1} \cdot \sum_{i=1}^n (x_{i1} - \hat{\mu}_1) \cdot (x_{i2} - \hat{\mu}_2)$$

- Statistics: **Multi-dimensional mean**
- Estimated **mean** of entire (numerical) data set.

mean of each col. $\rightarrow (1.3 \ 19.8 \ 5.4)$

- Estimated standard deviation $\hat{\sigma}_j = \sqrt{\hat{\sigma}_j^2}$
- Mean centering $x'_{ij} = x_{ij} - \hat{\mu}_j$

- Higher **variance** mean data points are more spread out.
- Lower **variance** mean the data points are closer to the mean, indicating less variability (or consistency)

- Positive **covariance** (> 0): Indicates that two variables tend to increase or decrease together.
- Negative **covariance** (< 0): Indicates that when one variable increases, the other tends to decrease.
- For example, outdoor temperature and heating cost often have negative **covariance**.
- Zero **covariance** ($= 0$): Indicates no linear relationship between the variables

$$D = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{matrix} & \begin{pmatrix} 1.3 & 19.8 & 5.4 \\ 2.3 & 1.4 & 3.4 \\ 2.8 & 4.7 & 1.2 \\ 5.6 & 2 & 9.1 \\ -0.5 & 9.4 & 2.3 \\ 0.4 & 1.8 & 6.4 \\ 1.1 & 2.1 & 3.5 \end{pmatrix} \end{matrix}$$

Covariance matrix $\Sigma = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \hat{\sigma}_{13} \\ \hat{\sigma}_{21} & \hat{\sigma}_2^2 & \hat{\sigma}_{23} \\ \hat{\sigma}_{31} & \hat{\sigma}_{32} & \hat{\sigma}_3^2 \end{pmatrix}$

Linear Algebra

- $\|x_i\|_2 = \sqrt{\sum_{k=1}^m x_{ik}^2}$ $\|x_i - x_j\|_2 = \sqrt{\sum_{k=1}^m (x_{ik} - x_{jk})^2} = \sqrt{(x_{11} - x_{21})^2 + (x_{12} - x_{22})^2 + (x_{13} - x_{23})^2}$
- $\|x\|_1 = \sum_{k=1}^m |x_k| = (|x_1| + |x_2| + \dots + |x_m|)^1$ $\|x_i - x_j\|_1 = \sum_{k=1}^m |x_{ik} - x_{jk}| = \sum_{k=1}^m |x_{ik} - x_{jk}| = \sum_{k=1}^3 |x_{1k} - x_{2k}| = |x_{11} - x_{21}| + |x_{12} - x_{22}| + |x_{13} - x_{23}|$
- $\|x\|_p = \sqrt[p]{\sum_{k=1}^m |x_k|^p} = (|x_1|^p + |x_2|^p + \dots + |x_m|^p)^{\frac{1}{p}}$ L_p norm of two vectors x_i, x_j with m dimensions is defined as follows:
- This is defined for any integer p greater than 0. $\|x_i - x_j\| = \sqrt[p]{\sum_{k=1}^m |x_{ik} - x_{jk}|^p} = \sqrt[4]{|0.2 - 0.4|^4 + |23 - 1|^4 + |5.7 - 5.4|^4}$

27. In data preprocessing, mean-centering is often applied to numerical features. Which of the following statements about mean-centering are true?
- Mean-centering involves subtracting the mean of each feature from its values.
 - Mean-centering ensures that each feature has a mean of zero.
 - Mean-centering is required for algorithms that are sensitive to scale, like k-NN. $\times$ standard normalization is required
 - Mean-centering helps reveal the direction of maximum variance in PCA.
 - Without mean-centering, covariance calculations may be biased by the original means.

Dot product $a^T b = \sum_{i=1}^m a_i b_i$ $x_3 \cdot x_5 = \sum_{k=1}^3 x_{3k} x_{5k} = x_{31} x_{51} + x_{32} x_{52} + x_{33} x_{53}$

- Cosine of the angle θ between two vectors
- a, b is calculated as follows:

$$\cos \theta = \frac{a \cdot b}{\|a\|_2 \|b\|_2}$$

	x_1	x_2	x_3
x_1	0.2	23	5.7
x_2	0.4	1	5.4
x_3	1.8	0.5	5.2
x_4	5.6	50	5.1
x_5	-0.5	34	5.3
x_6	0.4	19	5.4
x_7	1.1	11	5.5

$$\frac{x_2 \cdot x_3}{\|x_2\|_2 \|x_3\|_2} = \frac{(0.4 \ 1 \ 5.4)(1.8 \ 0.5 \ 5.2)}{\sqrt{0.4^2 + 1^2 + 5.4^2} \sqrt{1.8^2 + 0.5^2 + 5.2^2}} = 0.96$$

Integer encoding

A = 1
B = 2
C = 3

One-Hot encoding

	x_1	x_2
x_1	A	H
x_2	B	L
x_3	C	L
x_4	A	L
x_5	B	H
x_6	C	L
x_7	C	H
x_8	C	L

	x_{1A}	x_{1B}	x_{1C}	x_{2H}	x_{2L}
x_1	1	0	0	1	0
x_2	0	1	0	0	1
x_3	0	0	1	0	1
x_4	1	0	0	0	1
x_5	0	1	0	1	0
x_6	0	0	1	0	1
x_7	0	0	1	1	0
x_8	0	0	1	0	1

- Hamming Distance**: number of mismatches between two vectors

$$\delta_H(x_i, x_j) = \sum_{k=1}^m (x_{ik} \oplus x_{jk})$$

	x_1	x_2
x_1	A	H
x_2	B	L
x_3	C	L
x_4	A	L
x_5	B	H
x_6	C	L
x_7	C	H
x_8	C	L

	x_{1A}	x_{1B}	x_{1C}	x_{2H}	x_{2L}
x_1	1	0	0	1	0
x_2	0	1	0	0	1
x_3	0	0	1	0	1
x_4	1	0	0	0	1
x_5	0	1	0	1	0
x_6	0	0	1	0	1
x_7	0	0	1	1	0
x_8	0	0	1	0	1

$$\delta_H(x_1, x_2) = (1 \oplus 0) + (0 \oplus 1) + (0 \oplus 0) + (1 \oplus 0) + (0 \oplus 1) = 4$$

- We can also look at the Precision, Recall and F-score of the output as well which are contingency table-based methods.

$$prec_i = \frac{n_{ii}}{m_i} \quad recall_i = \frac{n_{ii}}{n_i} \quad F_i = \frac{2 \cdot prec_i \cdot recall_i}{prec_i + recall_i} \quad F = \frac{1}{k} \sum_{i=1}^k F_i$$

	T_1	T_2
C_1	5	0
C_2	1	9
C_3	0	13

- a. What is the precision of C_2 ?

$$prec(C_2) = \frac{5}{5}$$

- b. What is recall of C_3 ?

$$recall(C_3) = \frac{13}{9+13}$$

- c. What is F-score of C_1 ?

$$recall(C_1) = \frac{5}{6} \quad prec(C_1) = \frac{5}{5}$$

$$F\text{-score}(C_1) = \frac{2 \cdot \frac{5}{6} \cdot 1}{1 + \frac{5}{6}} = \frac{10 \cdot \frac{6}{6}}{6 + 11} = \frac{10}{17}$$

- The **clustering coefficient** of a node v_i is a measure of the density of edges in the neighborhood of v_i .
- Let $G_i = (V_i, E_i)$ be the subgraph induced by the neighbors of vertex v_i . Note that $v_i \notin V_i$, since we assume that G is simple. Let $|V_i| = n_i$ be the number of neighbors of v_i , and $|E_i| = m_i$ be the number of edges among the neighbors of v_i . The clustering coefficient of v_i is defined as:

$$C(v_i) = \frac{\# \text{ of edges in } G_i}{\text{maximum number of edges in } G_i} = \frac{m_i}{\binom{n_i}{2}} = \frac{2 \cdot m_i}{n_i(n_i - 1)}$$

Real World properties

Clustering effect

- Two nodes are more likely to be connected if they **share a common neighbor**.
- $C(k)$ denote the average clustering coefficient for all nodes with degree k .
- clustering effect also manifests itself as a power-law relationship between $C(k)$ and k :

$$C(k) \propto k^{-\gamma}$$

- log-log plot of k versus $C(k)$ exhibits a straight-line behavior with negative slope $-\gamma$
- High-degree nodes (hubs)** connect to many nodes, but their neighbors are often not connected to each other.
 - A **hub** connects to many **low-degree nodes** from different parts of the network.
 - These **low-degree nodes** are unlikely to be connected to each other.
- Low-degree nodes** are often in **tightly connected local communities**.
 - A low-degree node may **connect to only a few nodes** in a **well-defined cluster**.
 - These neighbors are more likely to be connected to each other, forming triangles.

Erdős-Rényi (ER) random graphs exhibit the small-world property

- The **closeness centrality** uses the sum of all the distances to rank how central a node is.

$$c(v_i) = \frac{1}{\sum_j d(v_i, v_j)}$$

Sum of distances from v_i to every node in network

Small-world property

- There is a **short path between any pair of nodes**.
- The graph G exhibits small-world behavior if the average path length μ_L scales logarithmically with the number of nodes in the graph, i.e.,

Scale-free property

- In many real-world graphs:
- the empirical degree distribution $f(k)$ exhibits a **scale-free** behavior captured by a **power-law** relationship with k , i.e., the probability that a node has degree k satisfies the condition:
- a power-law indicates that the vast majority of nodes have very small degrees.
- Few "Hub" nodes have high degree—they connect to or interact with lot of nodes.

The Barabási-Albert (BA) random graph model primarily exhibits the **scale-free property**, characterized by a power-law degree distribution where a few nodes (hubs)

Eccentricity Centrality

- Eccentricity** centrality is thus defined as follows

$$c(v_i) = \frac{1}{e(v_i)} = \frac{1}{\max_j \{d(v_i, v_j)\}}$$

- The less eccentric a node is the more central it is.

- The **eccentricity** of a node v_i is the maximum distance from v_i to any other node in the graph:

$$e(v_i) = \max_j \{d(v_i, v_j)\}$$

Matrix vector multiplication

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 4 \\ -4 & 1 & 2 \end{pmatrix} \quad v = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

For $A \in \mathbb{R}^{m \times n}$, $v \in \mathbb{R}^n$, then the i th element of Av is: $(Av)_i = \sum_{j=1}^n A_{ij}v_j$

$$Av = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 4 \\ -4 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1(1) + 3(1) + 0(2) \\ 0(1) + 2(1) + 4(2) \\ -4(1) + 1(1) + 2(2) \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 1 \end{pmatrix}$$

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$$S = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \quad R = \begin{pmatrix} \cos \frac{\pi}{6} & -\sin \frac{\pi}{6} \\ \sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{pmatrix} \quad x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$R \begin{pmatrix} 4 \cos \frac{\pi}{6} & -4 \sin \frac{\pi}{6} \\ 1 \sin \frac{\pi}{6} & 2 \cos \frac{\pi}{6} \end{pmatrix} = \begin{pmatrix} 4 \cos \frac{\pi}{6} & -4 \sin \frac{\pi}{6} \\ 1 \sin \frac{\pi}{6} & 2 \cos \frac{\pi}{6} \end{pmatrix}$$

Simplify this.

What happens in high dimensions?

- Let's do a in-class activity: Compute the volume of a **sphere** with radius 1 in 4 dimensions and 12 dimensions.

$$\text{vol}(S_d(r)) = K_d(r^d) = \left(\frac{\pi^{d/2}}{\Gamma(\frac{d}{2} + 1)} \right) r^d \quad \Gamma\left(\frac{d}{2} + 1\right) = \begin{cases} \left(\frac{d}{2}\right)! & \text{if } d \text{ is even} \\ \sqrt{\pi} \left(\frac{d-1}{2}\right)! & \text{if } d \text{ is odd} \end{cases}$$

$$\Rightarrow \left(\frac{\pi^2}{\Gamma(\frac{4}{2} + 1)} \right) 1^4 = \left(\frac{\pi^2}{\Gamma(3)} \right) 1 = \left(\frac{\pi^2}{2!} \right) = \left(\frac{\pi^2}{2} \right) = 4.934802 \quad (d = 4)$$

$$\Rightarrow \left(\frac{\pi^{12}}{\Gamma(\frac{12}{2} + 1)} \right) 1^{12} = \left(\frac{\pi^6}{\Gamma(7)} \right) 1 = \left(\frac{\pi^6}{6!} \right) = \left(\frac{\pi^6}{720} \right) = 1.335263 \quad (d = 12)$$

What happens in high dimensions?

- It gets weird when it goes to higher dimensions.
- Our intuition for 2- and 3-dimensional space doesn't always help us analyze data in high dimensions.
- Example: Volume of a **sphere**.
 - In 2-dimensions, the volume of a **sphere** is $\frac{\pi r^2}{2}$.
 - In 3-dimensions, the volume of a **sphere** is $\frac{4}{3}\pi r^3$.
 - In d-dimensions, the volume of a **sphere** is $\Gamma\left(\frac{d}{2} + 1\right) = \begin{cases} \left(\frac{d}{2}\right)! & \text{if } d \text{ is even} \\ \sqrt{\pi} \left(\frac{d-1}{2}\right)! & \text{if } d \text{ is odd} \end{cases}$

As dimensionality increases, the volume increases up to a point then decreases again.

Association Rules

- Support for the rule is the number of transactions in which both X and Y occur as subsets.
 - $s = \text{Sup}(X \rightarrow Y) = |t(XY)| = \text{Sup}(XY)$
- relative support** is defined as the fraction of transactions where X and Y co-occur.
 - $\text{rsup}(X \rightarrow Y) = \frac{\text{Sup}(XY)}{|D|} = P(X \wedge Y) = P(XY)$
- Confidence of the rule is the conditional probability that transaction contains Y given that it contains X .
 - $c = \text{Conf}(X \rightarrow Y) = P(Y|X) = \frac{P(XY)}{P(X)} = \frac{\text{rsup}(XY)}{\text{rsup}(X)} = \frac{\text{Sup}(XY)}{\text{Sup}(X)}$
- lift** is defined as the ratio of the observed joint probability of X and Y to the expected joint probability if they were statistically independent
 - $\text{lift}(X \rightarrow Y) = \frac{P(XY)}{P(X)P(Y)} = \frac{\text{conf}(X \rightarrow Y)}{\text{rsup}(Y)}$
 - Lift 1 means No association, >1 means positive association and <1 means negative association.

itemsets, with enough support and enough confidence?

Transaction ID	Items
1	Toilet paper(9), beans(2), rice(8), milk(5), baby wipes(1), diapers(4)
2	Oat milk(6), beans(2), toilet paper(9), orange juice(7)
3	Oat milk(6), milk(5), orange juice(7), toilet paper(9)
4	Beans(2), Toilet paper(9), baby wipes(1), diapers(4)
5	Toilet paper(9), butter(3), baby wipes(1), diapers(4)
6	Milk(5), Toilet paper(9)
7	Milk(5), rice(8)
8	Beans(2), Milk(5), Rice(8), Toilet Paper(9)
9	Milk(5), butter(3), diapers(4)
10	Beans(2), rice(8), Toilet paper(9)

Frequent item sets of size 1: {1}, {2}, {4}, {5}, {8}, {9}

Frequent item sets of size 2: {1,4}, {1,9}, {2,8}, {2,9}, {4,9}, {5,8}, {5,9}, {8,9}

Frequent item sets of size 3: {1,4,9}, {2,8,9}
Example: {4} $\rightarrow$ {1}

{4} has support 4 and {1,4} has support 3

So, we say that the rule {4} $\rightarrow$ {1} has support 3 and confidence $\frac{3}{4} = 0.75$

$$\text{Sup}(\{4\} \rightarrow \{1\}) = \text{Sup}(\{1,4\}) = 3$$

$$\text{conf}(\{4\} \rightarrow \{1\}) = \frac{\text{Sup}(\{1,4\})}{\text{Sup}(\{4\})} = \frac{3}{4}$$

$$\text{lift}(\{4\} \rightarrow \{1\}) = \frac{\text{conf}(\{4\} \rightarrow \{1\})}{\text{rsup}(\{1\})} = \frac{\frac{3}{4}}{\left(\frac{3}{10}\right)} = \frac{10}{4}$$

Bayes Classifier

- You are given a training dataset D of n points.
 - $D = \{x_1, x_2, \dots, x_n\}$
- Each point $x_i \in D$ is in d dimensional space.
- You are also given class labels of each data point x_i , denoted as y_i .
 - $y = \{y_1, y_2, \dots, y_n\}$ this is the label column
 - $y_i \in \{c_1, c_2, \dots, c_k\}$ each label is from one of these k values
- The objective of the Bayes classifier is, given a new data point calculate the $p(c_i|x)$

Bayes Theorem

- What is the solution?
- We use an approach called **naïve** bayes.
- This is even simpler method.
- Surprisingly, this works very well for real world data.
- Assumption:
 - We assume that our attributes are all independent

28. Which of the following statements about **Bayes classifiers** and **Naive Bayes** are true? Circle all that apply.

- The Bayes classifier always assumes feature independence.
- Naive Bayes is a probabilistic classifier based on Bayes' Theorem with a strong independence assumption.
- The Bayes classifier gives the optimal prediction if the true underlying probability distribution is known.
- Naive Bayes can perform surprisingly well even when the independence assumption is violated.
- Naive Bayes computes the joint probability of the features directly from the training data.

No, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm does not require the number of clusters to be specified as a parameter. MathWorks +1

Instead, the algorithm automatically determines the number of clusters based on the density of the data. This is a major advantage over centroid-based algorithms like **K-Means**, which require a predefined number of clusters (k). MathWorks +3

- Using the PCA algorithm as described in lectures, what does the α parameter represent?
 - the largest eigenvalue of the covariance matrix
 - the minimum fraction of total variance to be preserved
 - the minimum number of principal components to use
 - the minimum number of new attributes to create.
- [2 points] Which of the following is a disadvantage of the PCA algorithm? Circle the answer.
 - PCA cannot project data onto nonlinear subspaces and thus fails to capture nonlinear relationships in the attributes of a data set.
 - PCA cannot be applied to data sets of very high dimensionality (e.g., 1000 or more attributes)
 - PCA can reduce the dimensionality of a data set to two or three dimensions, but not 4 or more.
 - The new attributes produced by PCA can be correlated with one another, making it difficult to determine which new attributes contribute most to the observed variance in the data.

K-Means Clustering - Finding Cluster 2

Setup:

- $\mu_1 = (3.9, 4)$ and $\mu_2 = (6.2, 6)$
- Each point is assigned to the nearest centroid using Euclidean distance

Calculating distances for each point:

For each point, I'll compute distance to μ_1 and μ_2 :

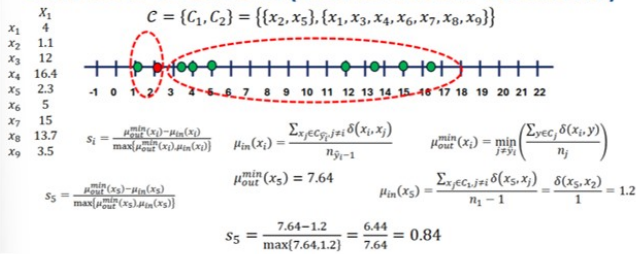
Point	(X_i, Y_i)	$d(\mu_1)$	$d(\mu_2)$	Assigned to
x_1	(5, 6)	$\sqrt{(1.2)^2 + 4} = 2.28$	$\sqrt{(1.44)^2 + 0} = 1.2$	C2
x_2	(4.9, 5.1)	$\sqrt{(1+1.2)^2} = 1.49$	$\sqrt{(1.69+0.81)} = 1.58$	C1

$x_1 = (5, 6)$ is the only point closer to $\mu_2 = (6.2, 6)$ than to $\mu_1 = (3.9, 4)$, because it sits right next to μ_2 on the X_2 axis.

26. What type of clustering method is k-means clustering?

- a representative-based method
- a density-based method
- a hierarchical method
- a graph-based method

Silhouette coefficient (Ground truth is not known)



Quick Template for Any Problem

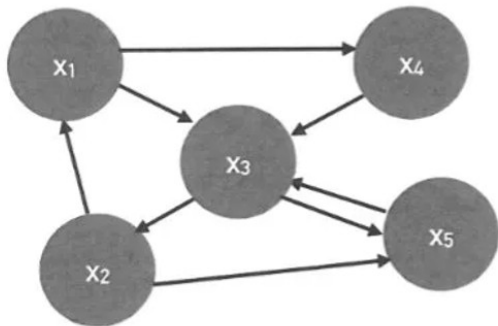
1. Write out A^T (transpose of adjacency matrix)
2. Start with $p_0 = [1, 1, 1, \dots, 1]^T$
3. Multiply: $\text{raw} = A^T \cdot p_{\text{current}}$
4. Normalize: $p_{\text{next}} = \text{raw} / \max(\text{raw})$
5. Repeat steps 3-4 for each iteration
6. The final vector = prestige scores

calculate the adjacency matrix A

$$A = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad A^T = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

$$A^T \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 1 \\ 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1/3 \\ 1/3 \\ 1 \\ 1/3 \\ 2/3 \end{pmatrix} = \begin{pmatrix} 0.33 \\ 0.33 \\ 1 \\ 0.33 \\ 0.66 \end{pmatrix} \leftarrow p_1$$

14. Consider the following graph.



Suppose we wish to find the prestige (eigenvector centrality) of each node in the network, and we are using Power Iteration. If we set the initial prestige vector p_0 to be the vector of 1s, what is the prestige vector going to be after the third iteration (what is $\frac{p_3}{\max(p_3)}$ going to be?)

$$A^T \begin{pmatrix} 0.33 \\ 0.33 \\ 1 \\ 0.33 \\ 0.66 \end{pmatrix} = \begin{pmatrix} 0.33 \\ 1.33 \\ 0.33 \\ 1.33 \\ 1.33 \end{pmatrix} \rightarrow \begin{pmatrix} 0.25 \\ 0.75 \\ 0.25 \\ 0.75 \\ 0.25 \end{pmatrix} \leftarrow p_2$$

$$A^T \begin{pmatrix} 0.25 \\ 0.75 \\ 0.25 \\ 0.75 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 0.75 \\ 1.5 \\ 0.25 \\ 1.75 \\ 1.75 \end{pmatrix} \rightarrow \begin{pmatrix} 0.42 \\ 0.57 \\ 0.86 \\ 0.14 \\ 1 \end{pmatrix} \leftarrow p_3$$

Step 1: Standard Normalize Each Feature (Column)

Formula: $z = (\text{value} - \text{mean}) / \text{std}$

Column X_1 :

Values: 0.2, 0.4, 1.8, 5.6, -0.5, 0.4, 1.1

Mean = $(0.2+0.4+1.8+5.6+0.5+0.4+1.1)/7 = 8.8/7 \approx 1.257$

Std = $\sqrt{\sum(x_i - \text{mean})^2/n} \approx 1.857$

Point	Raw	z-score
x_1	0.2	$(0.2-1.257)/1.857 = -0.569$
x_5	-0.5	$(-0.5-1.257)/1.857 = -0.946$

Column X_2 :

Values: 23, 1, 0.5, 50, 34, 19, 11

Mean = $138.5/7 \approx 19.786$

Std ≈ 16.517

Point	Raw	z-score
x_1	23	$(23-19.786)/16.517 \approx +0.195$
x_5	34	$(34-19.786)/16.517 \approx +0.861$

Column X_3 :

Values: 5.7, 5.4, 5.2, 5.1, 5.3, 5.4, 5.5

Mean = $37.6/7 \approx 5.371$

Std ≈ 0.185

Point	Raw	z-score
x_1	5.7	$(5.7-5.371)/0.185 \approx +1.778$
x_5	5.3	$(5.3-5.371)/0.185 \approx -0.384$

Step 2: Normalized Vectors

$$x_1 = (-0.569, 0.195, 1.778)$$

$$x_5 = (-0.946, 0.861, -0.384)$$

Step 3: Compute $\|x_1 - x_5\|^2$

$$x_1 - x_5 = (-0.569 - (-0.946), 0.195 - 0.861, 1.778 - (-0.384))$$

$$= (0.377, -0.666, 2.162)$$

$$\|x_1 - x_5\|^2 = (0.377)^2 + (-0.666)^2 + (2.162)^2$$

\$\$\$ 5.259 \$\$\$

Step 4: Apply Similarity Function ($\sigma = 1$)

$$\text{sim}(x_1, x_5) = e^{-\frac{\|x_1 - x_5\|^2}{2\sigma^2}} = e^{-\frac{5.259}{2}} = e^{-2.630}$$

$$\boxed{\text{sim}(x_1, x_5) \approx 0.072}$$

Interpretation

A similarity of 0.072 (close to 0) means x_1 and x_5 are quite dissimilar — mainly driven by their large difference in X_3 after normalization.

similarity between two features